

ELIJAH DICKINSON

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Education

Oklahoma State University

B.S. Aerospace Engineering & B.S. Mechanical Engineering; Minor in Mathematics

GPA: 3.43 | Extracurriculars: RUF, OSU Brazilian Jiu-Jitsu

Stillwater, OK

Aug. 2024 - May 2028

Engineering Projects

Mechanical Design Project | Project Planner, SolidWorks, 3D Printing, Arduino, C++

- Led **planning and task coordination** for a team-based **Arduino-controlled tank-drive robot**, translating project requirements into **milestones and assignments**.
- Designed mechanical components in **SolidWorks** and prepared models for **additive manufacturing** and system assembly.
- Manufactured and revised **3D-printed components** to improve **fit, alignment, and subsystem integration**.
- Programmed an **Arduino Uno in C++** and supported **mechanical-electrical integration** during system testing.
- Coordinated **CAD, fabrication, electronics, and programming** workstreams to keep subsystem assembly aligned with **team milestones**.

Wind Tunnel | Independent Engineering Project

- Designed a **benchtop subsonic wind tunnel** intended to measure **aerodynamic drag** on small test articles within a 6 x 6 in clear test section.
- Manufactured and sealed a **12 x 12 in inlet** and **12 x 12-to-6 x 6 in contraction section** by hand using wood panels, mechanical fasteners, and sealants.
- Integrated **honeycomb and dual-screen flow conditioning** into the inlet to reduce **large-scale flow disturbances** before contraction.
- Assembled and **leak-tested** the inlet-contraction interface, confirming **sealed joints** before test-section integration.
- Iterated panel geometry and alignment after identifying **out-of-square fit-up** and **dimensional tolerance issues**, improving section mating and seal quality.
- Fabricated **1/4 in acrylic test-section components** and evaluated **warping, fit-up, and alignment** during assembly.
- Specified an **Arduino Uno, HX711 amplifier, 5 kg load cell**, and **U-tube manometer** for planned drag and pressure measurements.
- Managed a **\$500 project budget** and invested **22.5+ hours** in independent engineering work, allocating **\$477** toward component sourcing, fabrication, assembly, design iteration, and testing.
- Documented dimensions, **bill of materials**, fabrication issues, design revisions, and test results in an engineering logbook.

Experience

LASSO Center

Certified Tutor

Jan. 2026 - Present

Stillwater, OK

- Delivered individualized tutoring in **Linear Algebra, Physics II, College Algebra, and Calculus I** by converting complex quantitative material into **structured problem-solving methods**.
- Diagnosed **conceptual gaps** and **adapted explanations**, examples, and practice problems to each student's needs.
- Developed **targeted practice problems** and step-by-step solution frameworks for **homework, quiz, and exam preparation**.
- Reinforced **independent problem-solving** by guiding students through **setup, algebra, units, and result verification**.

Casey's Convenience Store

Shift / Team Lead

Apr. 2022 - Present

Oklahoma

- Led **shift operations** in a high-volume retail environment, **delegated tasks**, maintained service standards, and coordinated **team workflow**.
- Resolved **customer concerns**, balanced competing priorities, and sustained efficient front-of-house operations during **peak periods**.
- Supported **cash handling**, stocking, food-service readiness, and store upkeep while reinforcing **team accountability**.

Wingstop

Team Member

Jan. 2025 - Present

Oklahoma

- Executed **high-volume order preparation and fulfillment** while maintaining **food quality, accuracy, and sanitation standards**.
- Coordinated with teammates **across stations** to sustain **throughput** and respond to changing order volume.

Whitmore Family Farms

Cashier

Aug. 2024 - Present

Oklahoma

- Managed **point-of-sale transactions** and **customer interactions** in a fast-paced farmers-market environment.
- Educated shoppers on **farm products and production practices** while supporting **weekly market operations**.

Technical Skills

CAD / Programming: SolidWorks | Python | C++ | Arduino Uno | LabVIEW | Microsoft Office Suite

Fabrication / Instrumentation: 3D Printing | Hand Fabrication | Drilling | Acrylic Scoring/Cutting | Mechanical Fastening | Sealing/Fit-Up | HX711 Load Cell Amplifier | U-Tube Manometry